

Electromagnetic Induction

CHAPTER SIX
CLASS - XII

MANISH JOSHI

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Learning Objectives:

- Electromagnetic induction

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- Electromagnetic induction
- Faraday's laws

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- Induced EMF and current

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- AC generator

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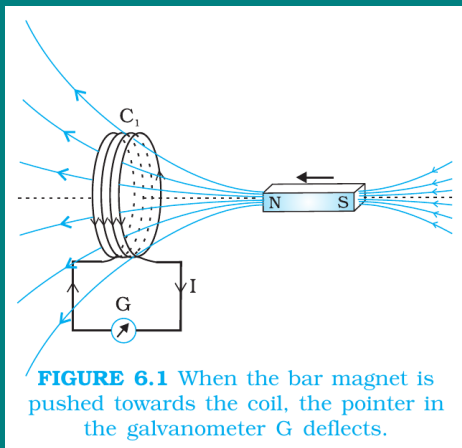
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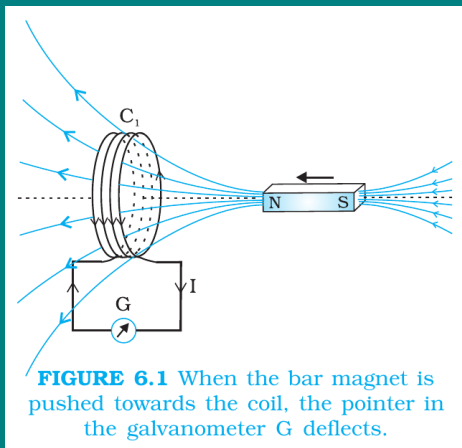
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"What is the use of a newborn baby?"

THE EXPERIMENTS OF FARADAY AND HENRY



Experiment 1: Magnet and Coil

THE EXPERIMENTS OF FARADAY AND HENRY



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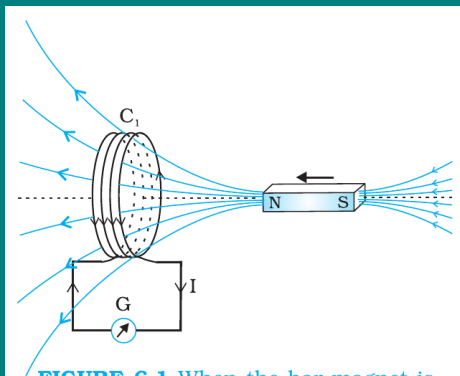
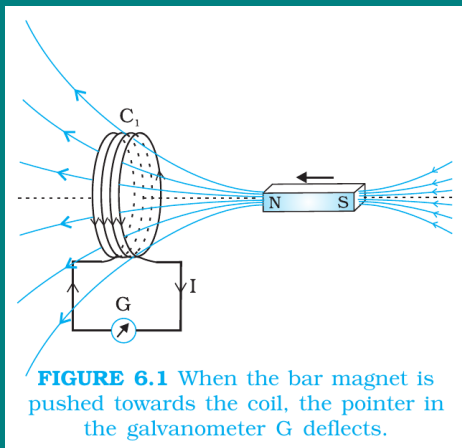


FIGURE 6.1 When the bar magnet is pushed towards the coil, the pointer in the galvanometer G deflects.

Experiment 1: Magnet and Coil

- A bar magnet moves relative to a coil connected to a galvanometer.
- Current is induced only during motion.

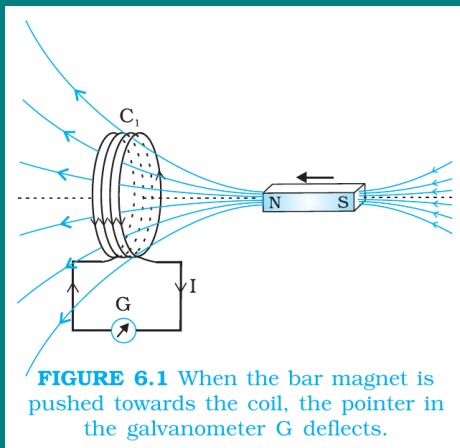
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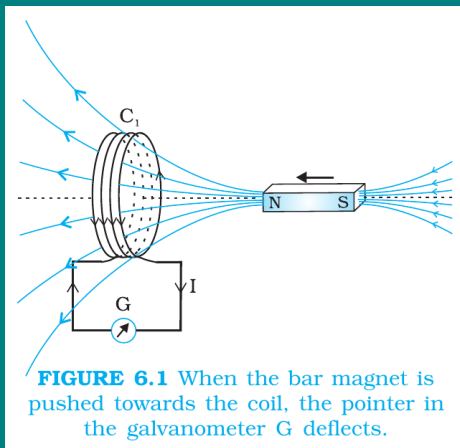
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- Key insight: **Relative motion** causes induction.

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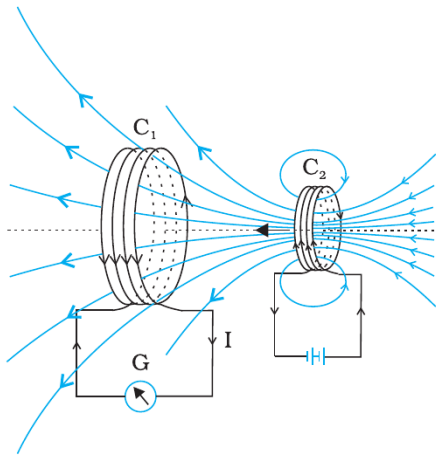


FIGURE 6.2 Current is

Experiment 2: Two Coils

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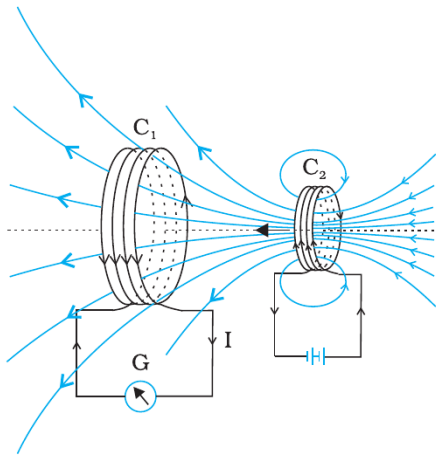


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- Coil C_2 carries a current and is moved near coil C_1 (with galvanometer).

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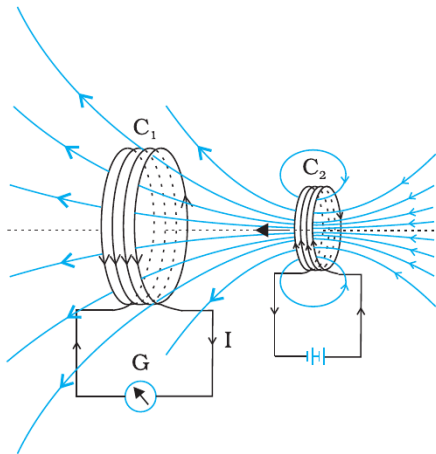


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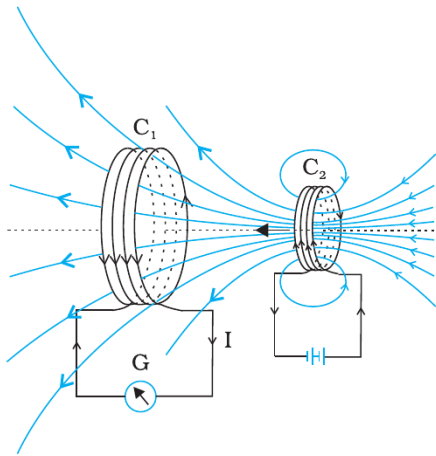


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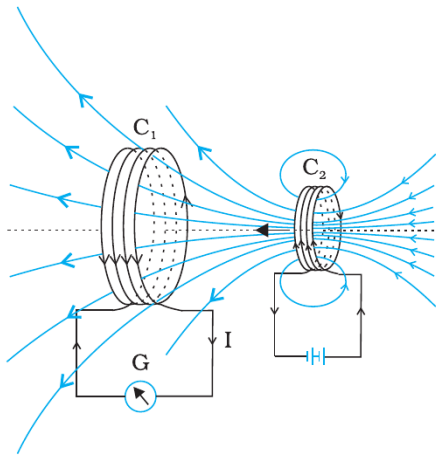


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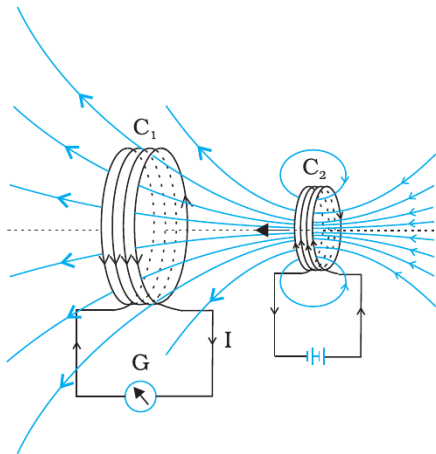
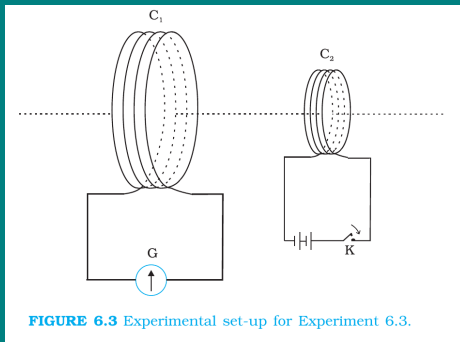


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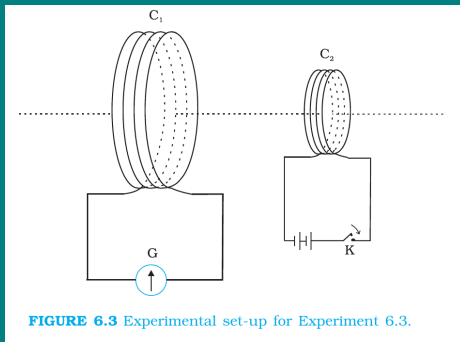
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- Induced current is observed in C_1 when:
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 - C_1 is moved relative to C_2 .
- Again, **relative motion** is essential.

THE EXPERIMENTS OF FARADAY AND HENRY



Experiment 3: Stationary Coils

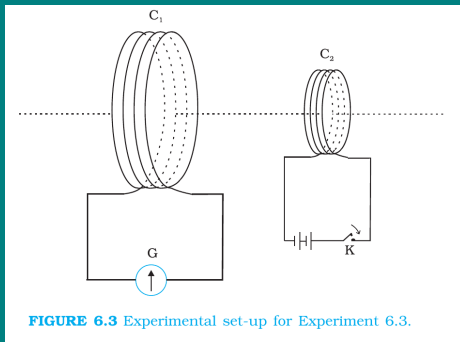
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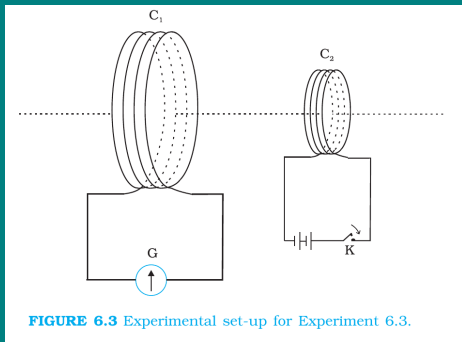
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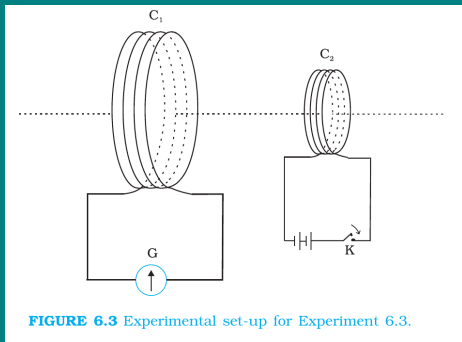
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Experiment 3: Stationary Coils

- Both coils are stationary.
- Coil C_2 is connected to a battery via a tapping key.
- **Momentary current is observed** in C_1 , when The **key is pressed or released**.
- **No motion needed** — only a **changing magnetic field**.

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Example 6.1 Consider Experiment 6.2. (a) What would you do to obtain a large deflection of the galvanometer? (b) How would you demonstrate the presence of an induced current in the absence of a galvanometer?

Solution

- (a) To obtain a large deflection, one or more of the following steps can be taken: (i) Use a rod made of soft iron inside the coil C_2 , (ii) Connect the coil to a powerful battery, and (iii) Move the arrangement rapidly towards the test coil C_1 .
- (b) Replace the galvanometer by a small bulb, the kind one finds in a small torch light. The relative motion between the two coils will cause the bulb to glow and thus demonstrate the presence of an induced current.

In experimental physics one must learn to innovate. Michael Faraday who is ranked as one of the best experimentalists ever, was legendary for his innovative skills.

EXAMPLE 6.2

Example 6.2 A square loop of side 10 cm and resistance $0.5\ \Omega$ is placed vertically in the east-west plane. A uniform magnetic field of 0.10 T is set up across the plane in the north-east direction. The magnetic field is decreased to zero in 0.70 s at a steady rate. Determine the magnitudes of induced emf and current during this time-interval.

Solution The angle θ made by the area vector of the coil with the magnetic field is 45° . From Eq. (6.1), the initial magnetic flux is

$$\begin{aligned}\Phi &= BA \cos \theta \\ &= \frac{0.1 \times 10^{-2}}{\sqrt{2}} \text{ Wb}\end{aligned}$$

Final flux, $\Phi_{\min} = 0$

The change in flux is brought about in 0.70 s. From Eq. (6.3), the magnitude of the induced emf is given by

$$\varepsilon = \frac{|\Delta \Phi_B|}{\Delta t} = \frac{|(\Phi - 0)|}{\Delta t} = \frac{10^{-3}}{\sqrt{2} \times 0.7} = 1.0 \text{ mV}$$

And the magnitude of the current is

$$I = \frac{\varepsilon}{R} = \frac{10^{-3} \text{ V}}{0.5 \Omega} = 2 \text{ mA}$$

Note that the earth's magnetic field also produces a flux through the loop. But it is a steady field (which does not change within the time span of the experiment) and hence does not induce any emf.

Example 6.3

A circular coil of radius 10 cm, 500 turns and resistance $2\ \Omega$ is placed with its plane perpendicular to the horizontal component of the earth's magnetic field. It is rotated about its vertical diameter through 180° in 0.25 s. Estimate the magnitudes of the emf and current induced in the coil. Horizontal component of the earth's magnetic field at the place is $3.0 \times 10^{-5}\text{ T}$.

Solution

Initial flux through the coil,

$$\begin{aligned}\Phi_{B(\text{initial})} &= BA \cos \theta \\ &= 3.0 \times 10^{-5} \times (\pi \times 10^{-2}) \times \cos 0^\circ \\ &= 3\pi \times 10^{-7} \text{ Wb}\end{aligned}$$

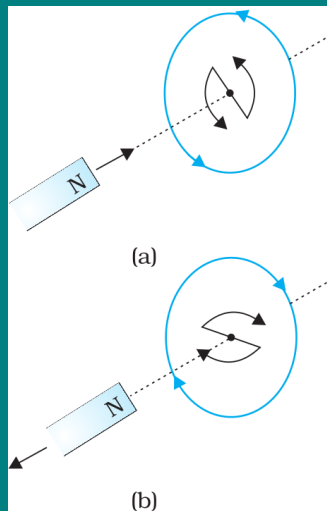
Final flux after the rotation,

$$\begin{aligned}\Phi_{B(\text{final})} &= 3.0 \times 10^{-5} \times (\pi \times 10^{-2}) \times \cos 180^\circ \\ &= -3\pi \times 10^{-7} \text{ Wb}\end{aligned}$$

Therefore, estimated value of the induced emf is,

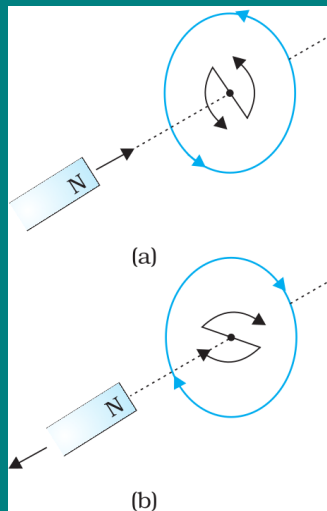
$$\begin{aligned}\varepsilon &= N \frac{\Delta \Phi}{\Delta t} \\ &= 500 \times (6\pi \times 10^{-7}) / 0.25 \\ &= 3.8 \times 10^{-3} \text{ V}\end{aligned}$$

Lenz's Law



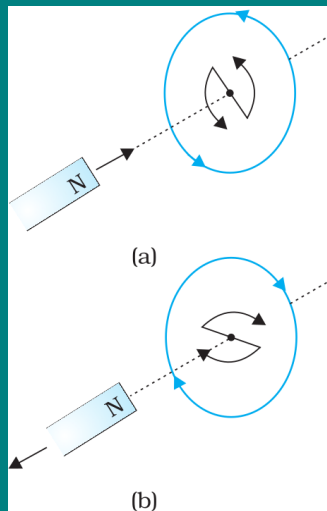
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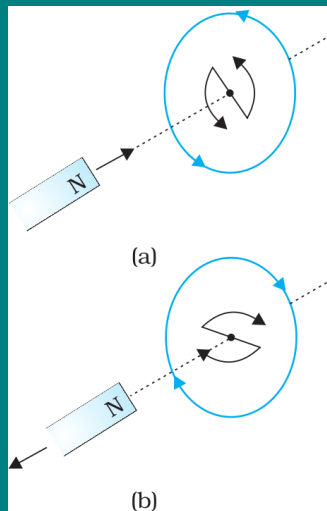
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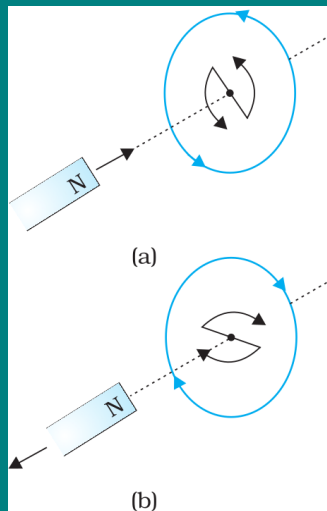
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- As the **North-pole** of the bar magnet **moves towards the coil**, the **magnetic flux** through the coil **increases**. Hence **current is induced** in the coil in such a **direction** that it **opposes the increase in flux**. This is **possible only** if the **current** in the coil is in a **counter-clockwise**.

- Conservation of Energy:

- Lenz's Law ensures the induced current does **not violate the law of conservation of energy**.
- To maintain motion (e.g., moving a magnet), **external work must be done**.
- This mechanical energy is converted into:
- **Electrical energy** (induced current), Then into **heat energy**.

- Why Lenz's Law is Necessary:

- If the induced current aided the change in flux:
It would result in continuous acceleration without input — a **perpetual motion machine**. This violates the **law of conservation of energy**.
- Lenz's Law prevents this by ensuring **energy balance**.

- Quick Example:

- Moving a bar magnet towards a loop → loop resists by creating opposing magnetic field → motion slows → **you must apply force** → energy is converted into **electric current** → dissipated as **heat**.

Example 6.4

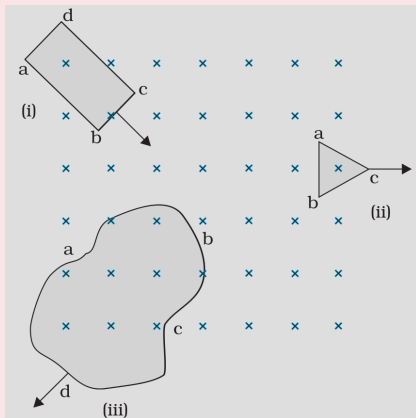


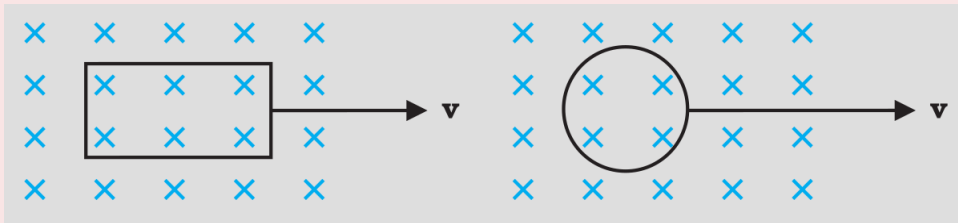
Figure 6.7 shows planar loops of different shapes moving out of or into a region of a magnetic field which is directed normal to the plane of the loop away from the reader. Determine the direction of induced current in each loop using Lenz's law.

Example 6.5

- (a) A closed loop is held stationary in the magnetic field between the north and south poles of two permanent magnets held fixed. Can we hope to generate current in the loop by using very strong magnets?
- (b) A closed loop moves normal to the constant electric field between the plates of a large capacitor. Is a current induced in the loop
 - (i) when it is wholly inside the region between the capacitor plates
 - (ii) when it is partially outside the plates of the capacitor? The electric field is normal to the plane of the loop.

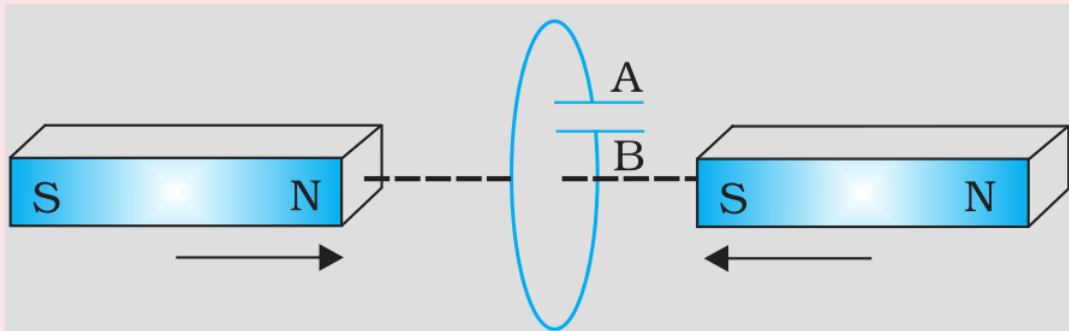
Example 6.5

- (c) A rectangular loop and a circular loop are moving out of a uniform magnetic field region (Fig. 6.8) to a field-free region with a constant velocity $\mathbf{v}$. In which loop do you expect the induced emf to be constant during the passage out of the field region? The field is normal to the loops.



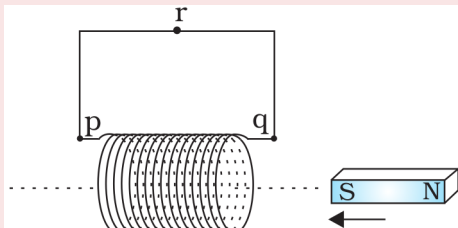
Example 6.5

(d) Predict the polarity of the capacitor in the situation described by Fig. 6.9.

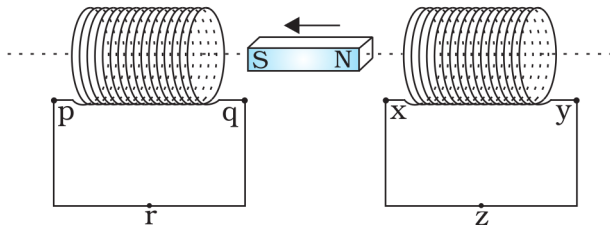


Exercise 6.1

Predict the direction of induced current in the situations described by the following Figs. 6.18(a) to (f).

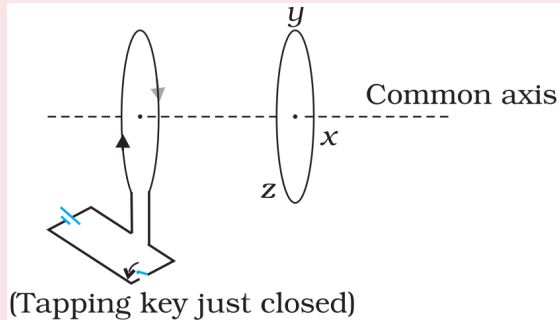


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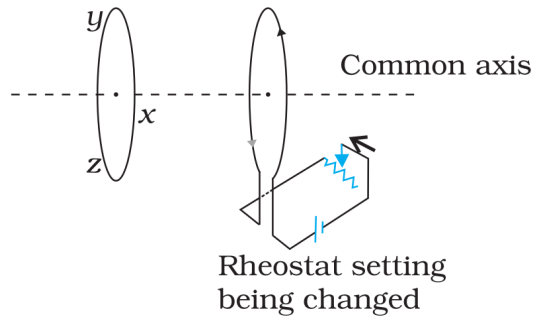


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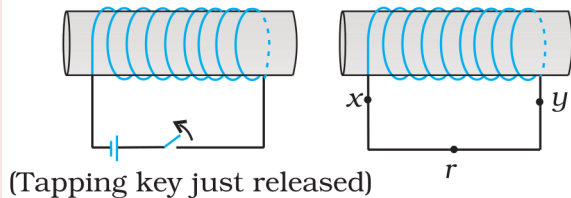


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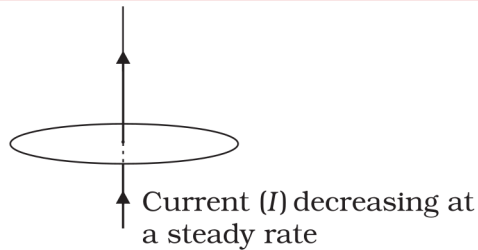


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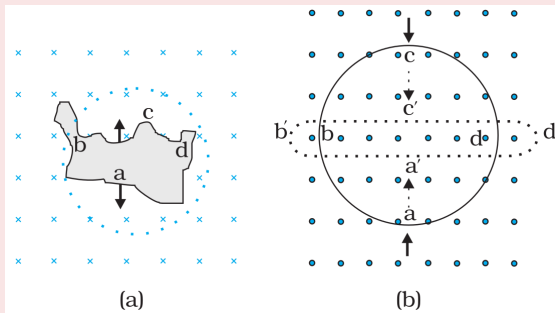
(e)



(f)

Exercise 6.2

Use Lenz's law to determine the direction of induced current in the situations described by Fig. 6.19: **(a)** A wire of irregular shape turning into a circular shape; **(b)** A circular loop being deformed into a narrow straight wire.

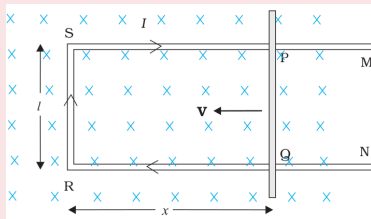


Exercise 6.3

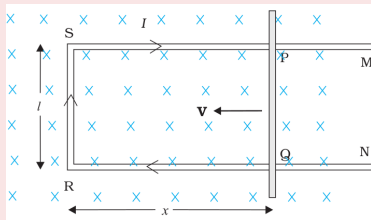
A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s, what is the induced emf in the loop while the current is changing?

Motional Electromotive Force

- A conducting rod **PQ** of length **l** is moving on a rail with **velocity v** in the **uniform magnetic field B** perpendicular to the plane of motion.

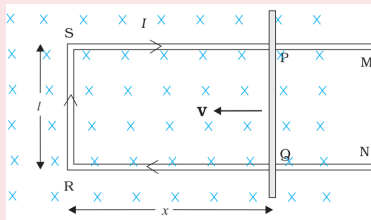


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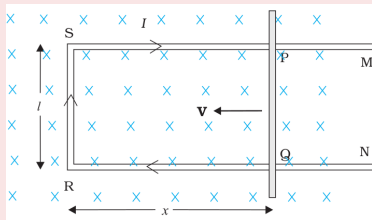
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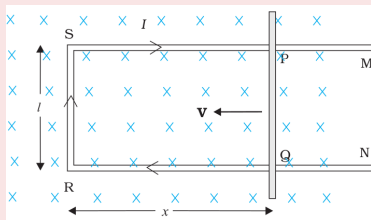
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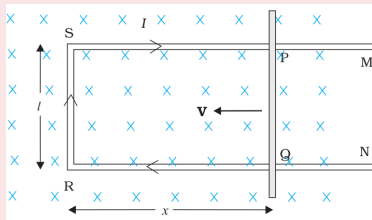
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$$\phi_B = B \cdot A = Blx$$

- Since **x is changing with time**, the **rate of change of flux ϕ_B** will **induce an emf** given by:

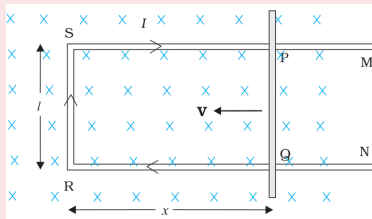
$$\epsilon = -\frac{d\phi_B}{dt} = -Bl\frac{dx}{dt}$$

Motional Electromotive Force



$$\epsilon = Blv \quad \left(\because \frac{dx}{dt} = v \right)$$

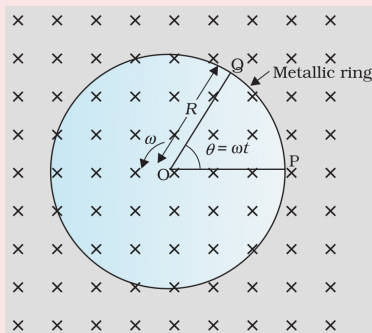
Motional Electromotive Force



$$\epsilon = Blv \quad \left(\because \frac{dx}{dt} = v \right)$$

- The induced emf Blv is called **motional emf**.

Example 6.6



A metallic rod of 1 m length is rotated with a frequency of 50 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring (Fig. 6.11). A constant and uniform magnetic field of 1 T parallel to the axis is present everywhere. What is the emf between the centre and the metallic ring?

Example 6.7

A wheel with 10 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of earth's magnetic field H_E at a place. If $H_E = 0.4 \text{ G}$ at the place, what is the induced emf between the axle and the rim of the wheel? Note that $1 \text{ G} = 10^{-4} \text{ T}$.

Exercise 6.4

A rectangular wire loop of sides 8 cm and 2 cm with a small cut is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the loop. What is the emf developed across the cut if the velocity of the loop is 1 cm s^{-1} in a direction normal to the (a) longer side, (b) shorter side of the loop? For how long does the induced voltage last in each case?

Exercise 6.5

A 1.0 m long metallic rod is rotated with an angular frequency of 400 rad s^{-1} about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant and uniform magnetic field of 0.5 T parallel to the axis exists everywhere. Calculate the emf developed between the centre and the ring.